

Exercise 2

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}^d$ and a given model $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}^d$ whose coordinates are $\hat{f}(x) = (\hat{f}_1(x), \dots, \hat{f}_d(x))$. Construct a score function $s(x, y)$ that would yield intervals of the form

$$\mathcal{C}(x) = \prod_{j=1}^d [\hat{f}_j(x) - c, \hat{f}_j(x) + c]$$

for a $c > 0$.

Solution Put

$$s(x, y) = \max_{j=1, \dots, d} |\hat{f}_j(x) - y_j|.$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$, we will have the desired interval with $c = \hat{q}$. ///